

Estimating an IS–LM / Mundell–Fleming System

with Real Data in Python

NAME AND SURNAME

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DATA: Quarterly macroeconomic data for the United States from the FRED database were used for the period 1990–2024. The variables included real GDP, the interest rate, government spending, the M2 money supply, the CPI price index, and the real exchange rate.

The original data were converted to quarterly frequency, and missing observations were removed. This resulted in a clean quarterly dataset for estimating the model.

Figure 1: $\log_y$

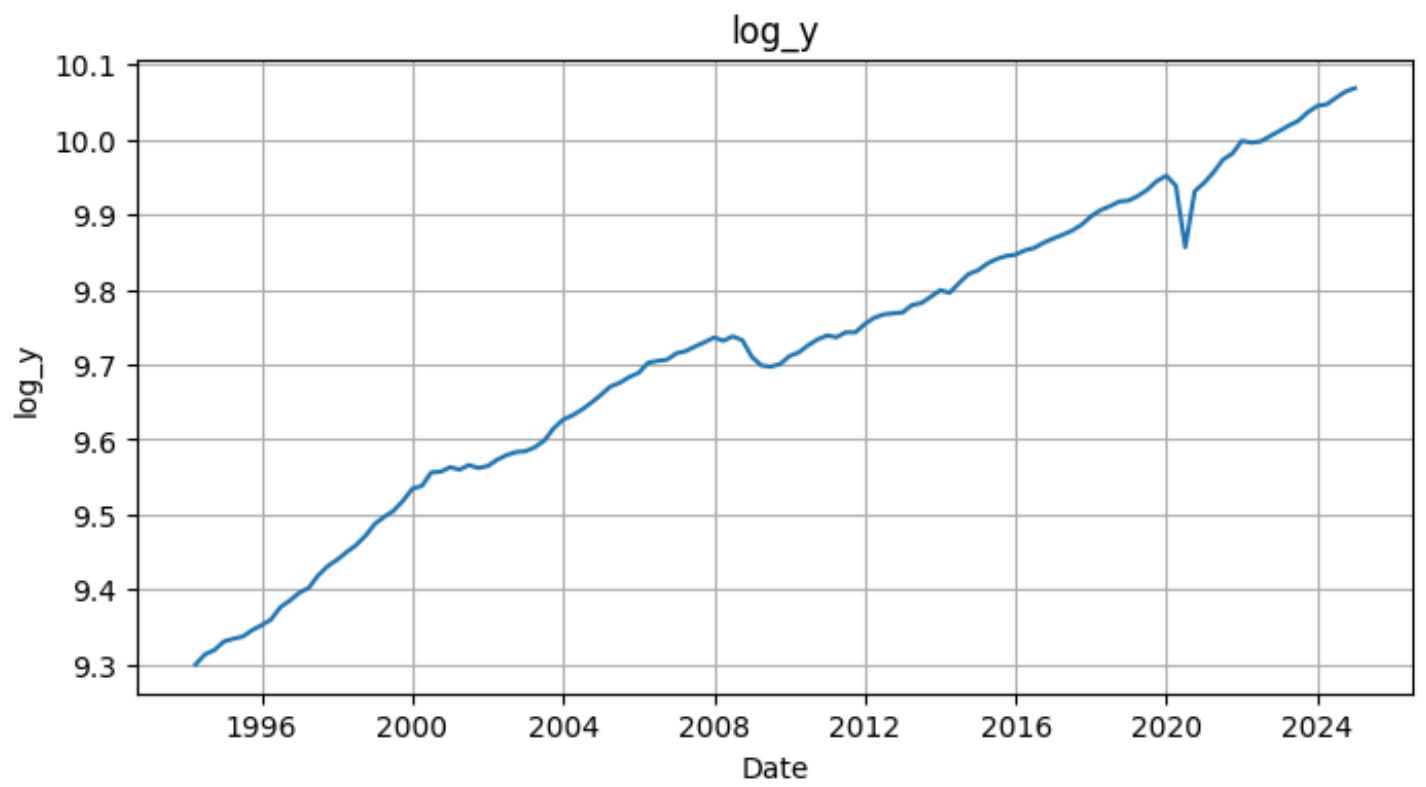


Figure 2: interest

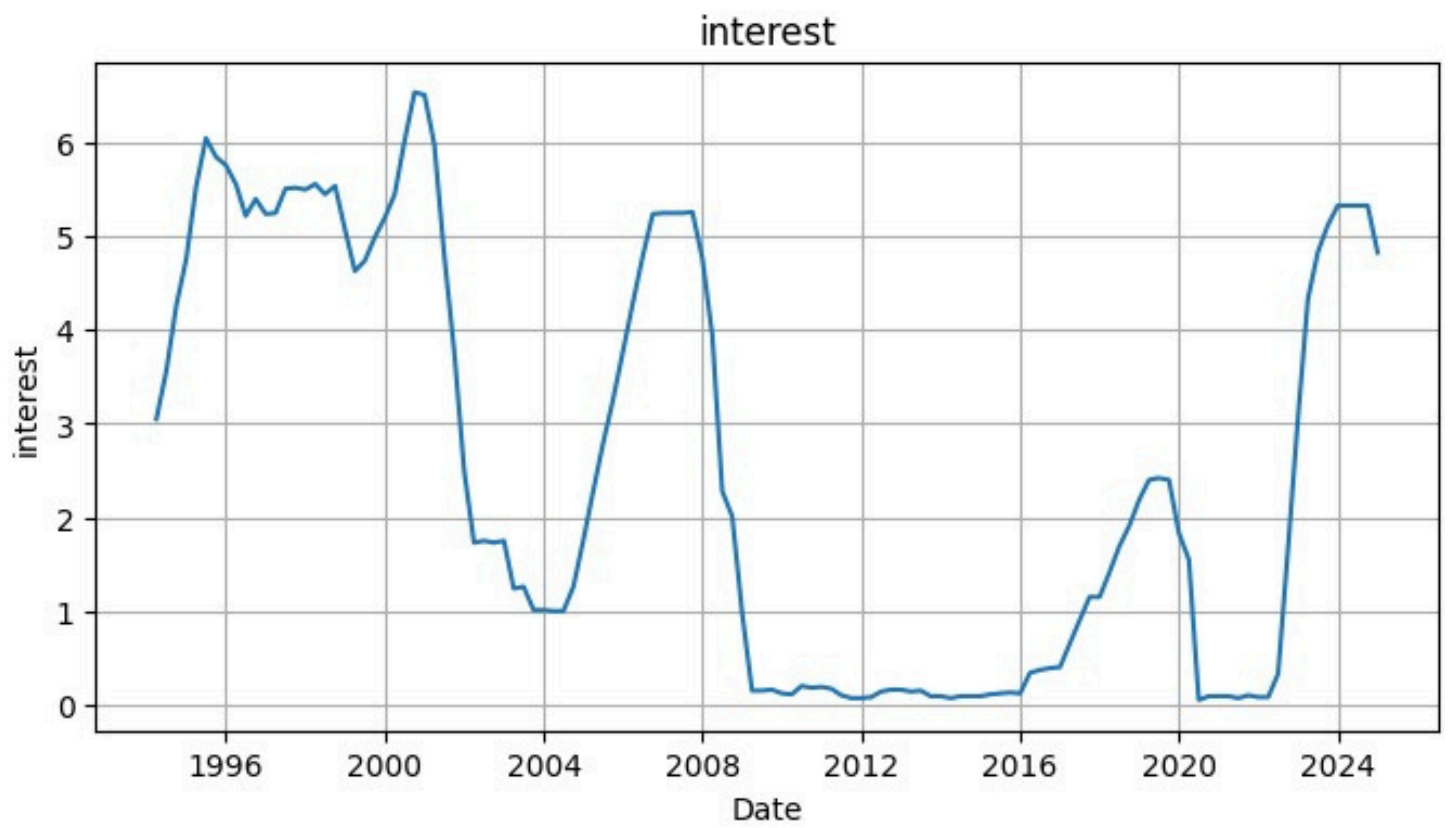


Figure 3: log_g

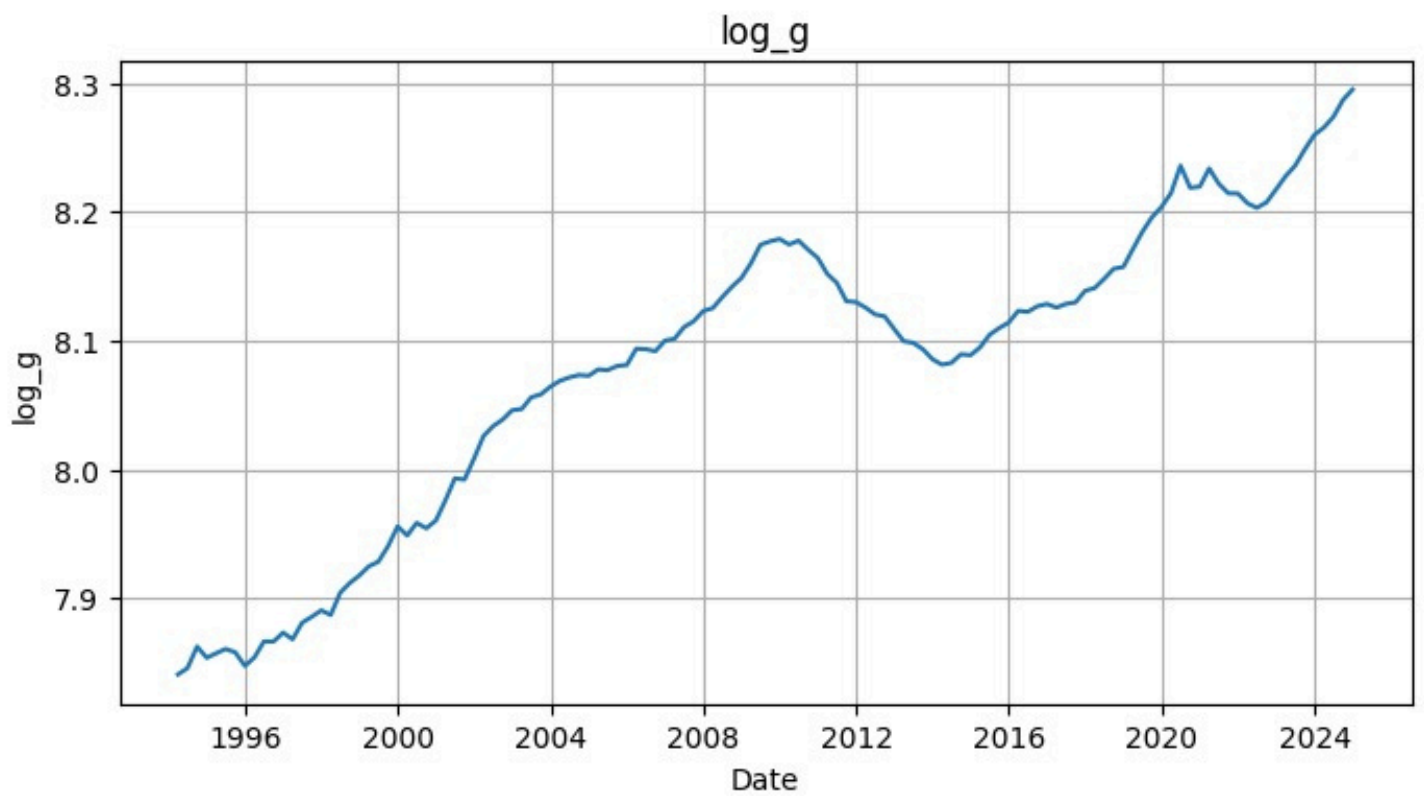


Figure 4: log_m_real

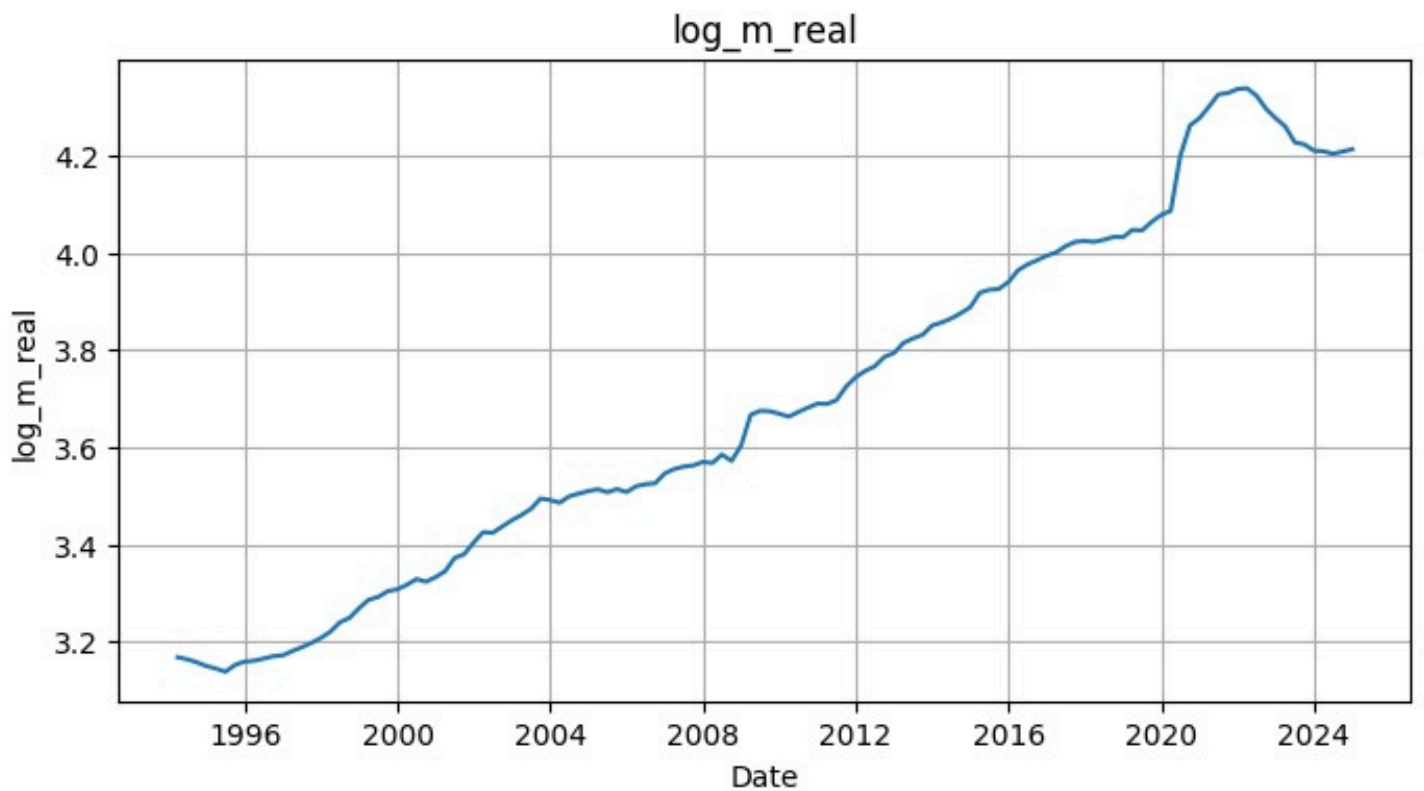
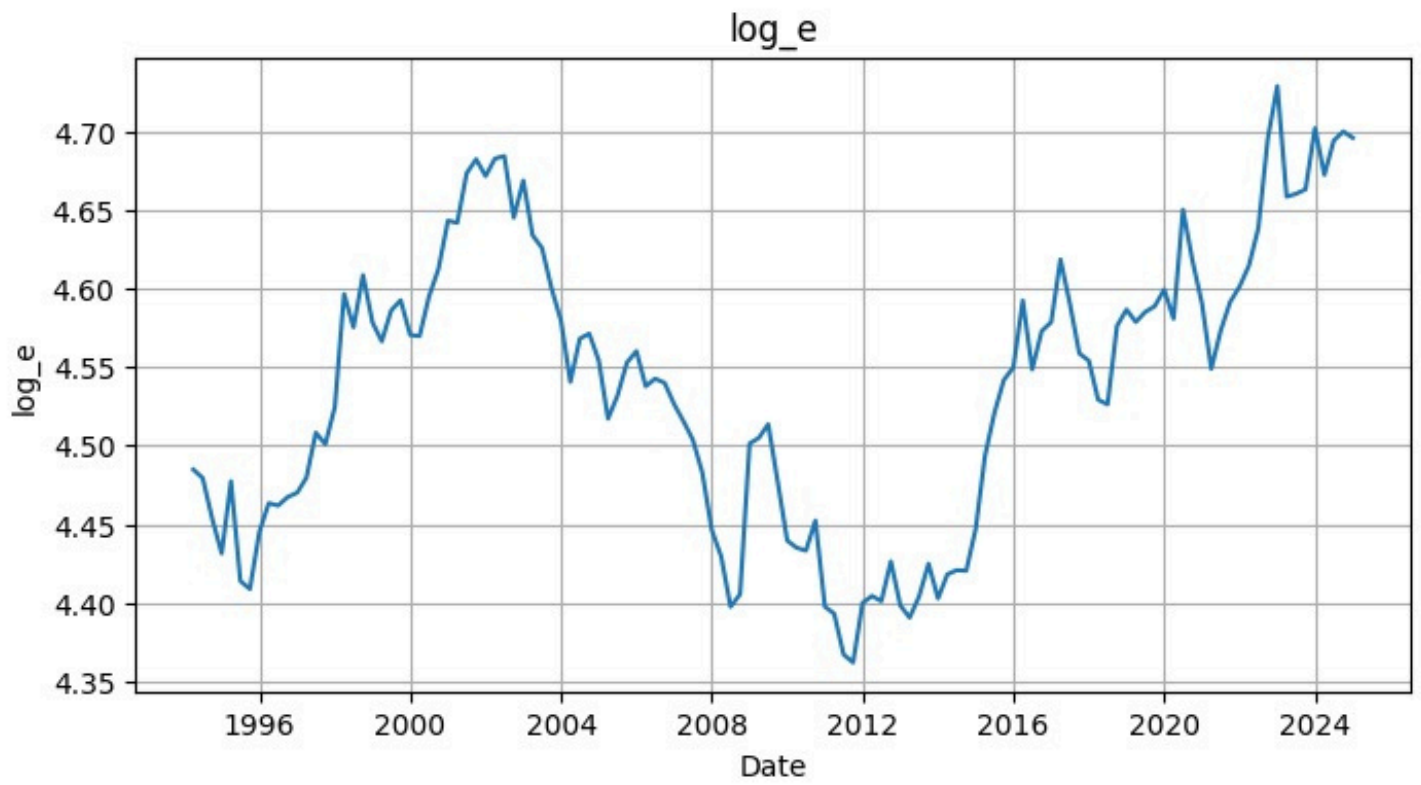
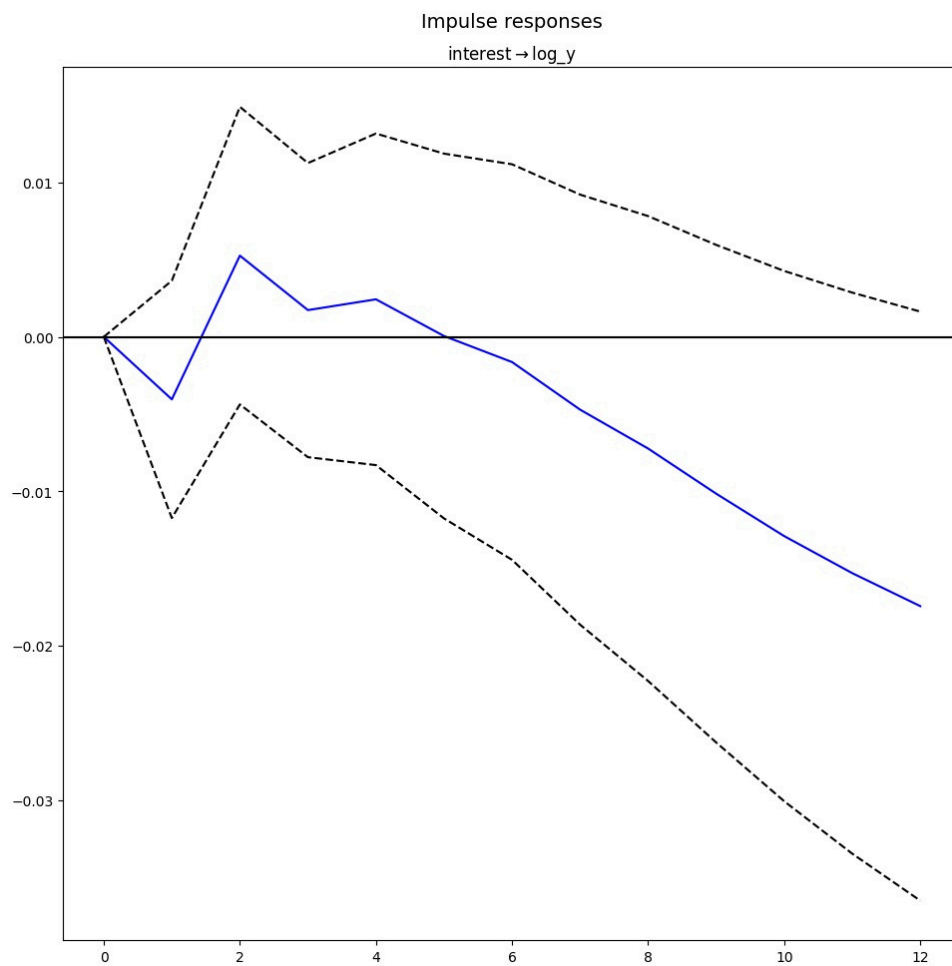


Figure 5: log_e



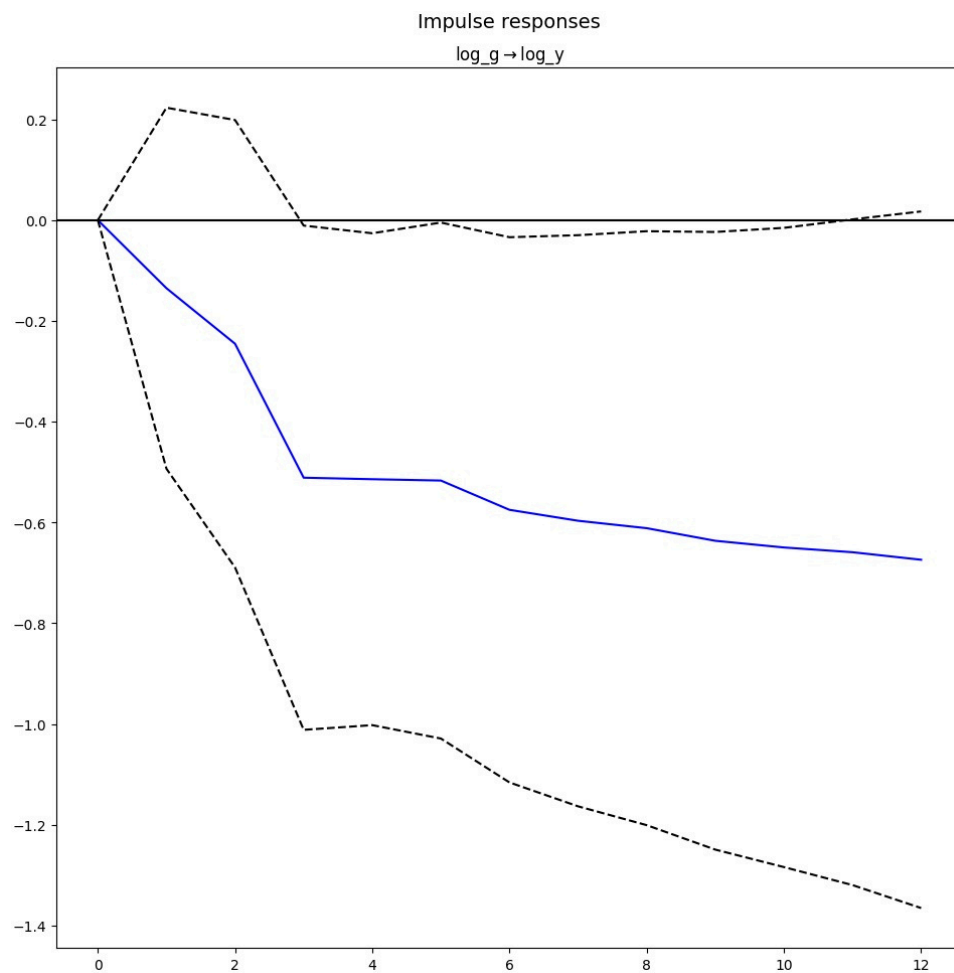
The graphs illustrate the evolution of the main macroeconomic variables over time.

Response of Output ($\log_y$) to an Interest Rate Shock



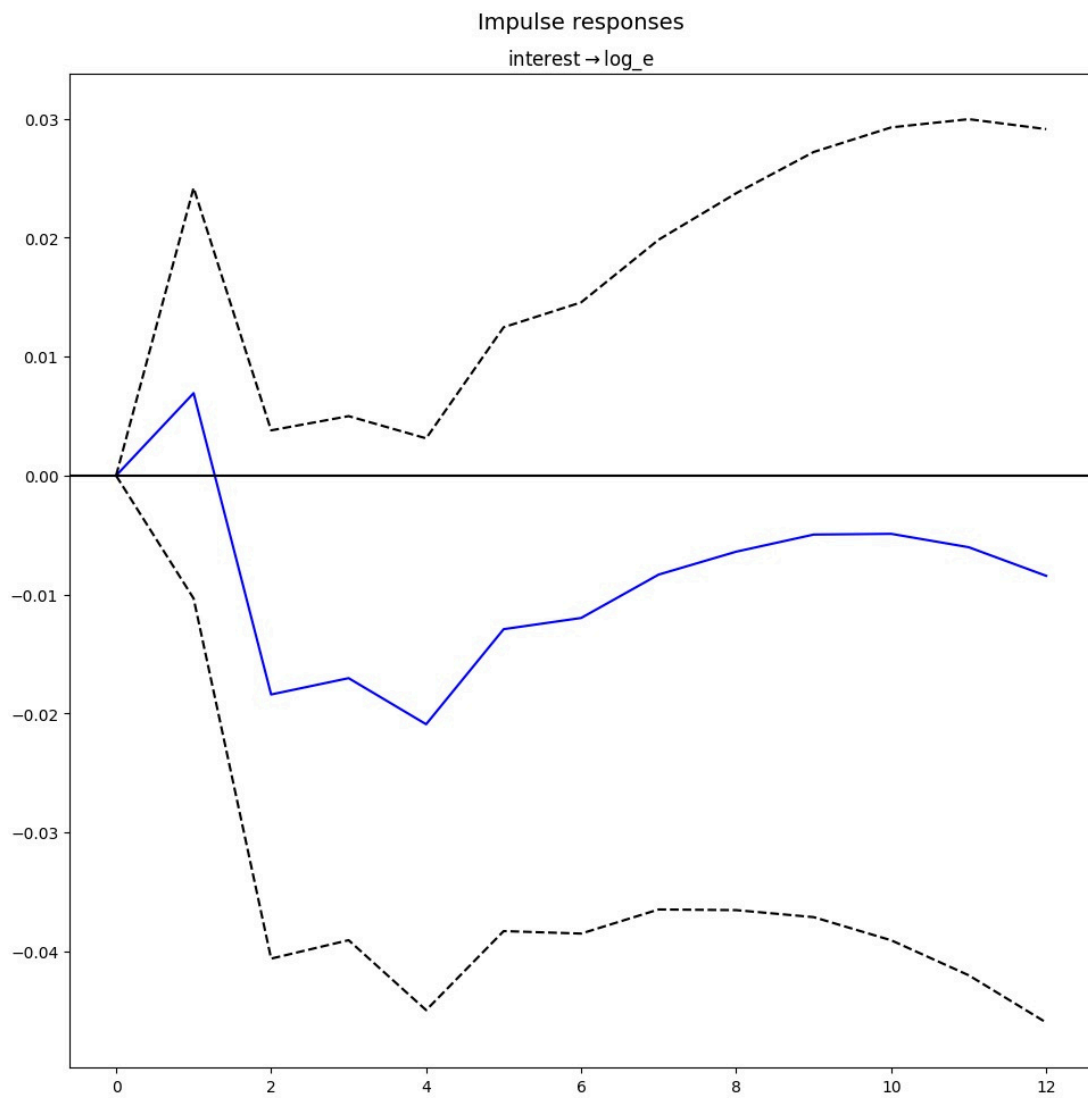
The graph shows how output (log_y) responds to an interest rate shock within the VAR model.

Response of Output (log_y) to a Government Spending Shock



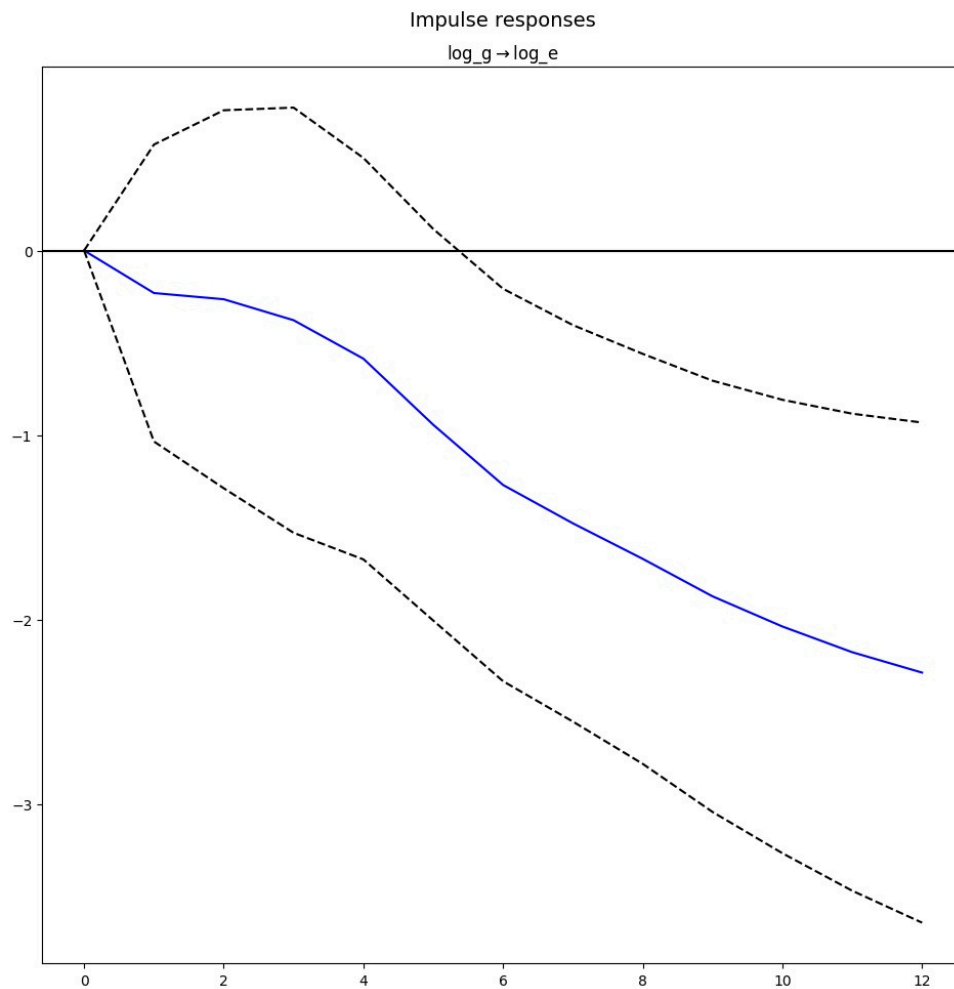
The graph shows how output ($\log_y$) responds to a government spending shock ($\log_g$) within the VAR model.

Response of Exchange Rate ($\log_e$) to an Interest Rate Shock



The graph shows how the exchange rate (log_e) responds to an interest rate shock within the VAR model.

Response of Exchange Rate (log_e) to a Government Spending Shock



The graph shows how the exchange rate (log_e) responds to a government spending shock (log_g) within the VAR model.

Economic Interpretation:

According to the IS curve, an increase in the interest rate reduces investment and output.

An increase in government spending shifts the IS curve to the right and increases output.

When the interest rate increases, domestic financial assets become more attractive, and the exchange rate may appreciate.

The empirical results are generally consistent with the theoretical predictions.

Connection to the Blanchard IS–LM / Mundell–Fleming Model:

The results are interpreted using the Blanchard IS–LM / Mundell–Fleming model. The theory predicts that an increase in the interest rate tends to reduce output, while an increase in government spending tends to stimulate economic activity. In an open economy, changes in the exchange rate also affect aggregate demand and output.

1. Does an interest rate shock reduce output?

According to the IS curve, an increase in the interest rate reduces investment and output. In an open economy, it may also affect the exchange rate and net exports.

2. Does government spending increase output?

An increase in government spending shifts the IS curve to the right and increases output. However, in an open economy, part of the increase in demand is directed toward imports, reducing the multiplier effect.

3. What happens to the exchange rate?

When the interest rate increases, domestic assets become more attractive, causing the exchange rate to rise and the domestic currency to appreciate.

Conclusion

The analysis confirms that the empirical relationships are broadly consistent with the theoretical predictions of the Blanchard IS–LM / Mundell–Fleming model. Interest rate shocks reduce output over time, while government spending supports economic activity. Overall, the project demonstrates how macroeconomic theory can be evaluated using real-world data and econometric techniques.

Additional Information

Conducted extensive research on macroeconomic theory and econometric analysis. AI-assisted tools were used to support research, improve documentation, and enhance the clarity of the final report.

Kostas Karamanis.